

# Curriculum Vitae

## Personal Details

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**Full Name:** Hamed Ghaderi  
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## Education

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**2022-2024**     **M.Sc in Gravitation and Cosmology**, University of Zanjan, Iran  
*Thesis: Classification of Galaxies from Morphological Structures Using Machine Learning*

**2016-2020**     **B.Sc in Computer Software Engineering**, Salman Higher Education Institute, Iran  
*Thesis: Development of a Python-Based Program for Deep Sky Astronomical Data Processing*

## Research Interests

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1. Observational Cosmology
2. Galaxy and Extra-Galaxy Data Analysis
3. ML and DL Cosmology
4. Big Surveys and Observational Instruments

## Publication

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1. **Ghaderi, H.**, Alipour, N., Safari, H., 2024 “Galaxy Classification via Machine Learning Approaches on the Zernike Moments and Images Features for GZ2 Catalog”. *Accepted to ApJS*. DOI: 10.48550/arXiv.2501.09816
2. Safari, H., Alipour, N., **Ghaderi, H.**, Garavand, P., 2023 “Zernike moments descript of solar and astronomical features: Python code”. *IJAA, Volume 10, Issue 3, pp. 267-268*. DOI: 10.22128/ijaa.2023.718.1162

## Skills and Expertise

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**Technical skills:** **Python Programming Language** (Advanced), **TensorFlow** (Advanced), **PyTorch** (Intermediate), **Scikitlearn** (Advanced), **Mathematica** (Intermediate), **CAMB** (Intermediate), **CosmoMC** (Beginner), **Topcat** (Beginner), **IRAF** (Intermediate), **Git** (Advanced), **Linux** (Advanced), **Adobe** (Advanced).

**Domain Expertise:** **Machine learning** (Advanced), **Computer Vision** (Advanced), **Neural Networks** (Advanced), **Astronomy and Cosmology** (Advanced), **Data Science** (Advanced), **Physics** (Advanced), **Astro Photography** (Advanced), **Hyper-lapse and Timelapse Videography** (Advanced), **Teaching** (Advanced), **Public Speaking** (Advanced).

## Research Experiences

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1. **Automated Galaxy Classification with Machine Learning and Deep Learning Algorithms**  
ML and computer vision algorithms / University of Zanjan, Iran     2022 – 2024
2. **Development of the ZEMO Python Package (A Feature Extraction tool)**  
Python programming language and Computer vision / University of Zanjan, Iran     2022-2023
3. **Development of a program for astronomical data analysis in Python**  
Python programming language / Salman Higher Education Institute, Iran     2019 - 2020
4. **Data Analysis on Near-Earth Objects and Main Belt Asteroids**  
Astrometrica Software / Data: Pan-Starrs Observatory     2018

5. **Photometric Analysis 40 Data of Galaxy M33**  
IRAF Data Analysis System (Scisoft, ESO) / Data: INT (Isaac Newton Telescope) 2014
6. **Photometric Analysis 100 Data of Centre of Galaxy M101**  
IRAF Data Analysis System (Scisoft, ESO) / Data: HST (Hubble Space Telescope) 2014

## Teaching Experiences

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1. University of Zanjan, Workshop on observation astronomy, Zanjan, Iran 2024
2. University of Zanjan, Fundamental of astronomy, Zanjan, Iran 2023
3. Khayyam University of Mashhad, Physics of Science Photography, Mashhad, Iran 2022
4. Ferdowsi Academy, Photography Workshops, Mashhad, Iran 2018
5. Khayyam University of Mashhad, Professional Astro Photometry, Mashhad, Iran 2014
6. Besat High School, Introductory and Special Subjects of Astronomy, Mashhad 2014-2015
7. Khayyam University of Mashhad, Observational Astronomy and Introduction of Photometry 2013-2015
8. Astronomical Centre of Qom, Stellar Astrophysics for Students Participating in the Olympiad of Astronomy 2009
9. Astronomical Centre of Qom, Dynamical Astronomy for Students Participating in the Olympiad of Astronomy 2009
10. Observatory of Astronomical Centre of Qom, Introduction of Observational Astronomy 2006-2008
11. Astronomical Centre of Qom, Introduction of Astronomy 2006-2007

## Certification

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1. **9<sup>th</sup> Byurakan International Summer School (9BISS)** September 2024  
Byurakan Astrophysical Observatory (BAO), Armenia ([Link](#))
2. **Workshop on Cosmology: from theory to observations** August 2023  
Institute for Research in Fundamental Sciences (IPM), Iran ([Link](#))
3. **16<sup>th</sup> National Meeting on Astronomy and Astrophysics (Oral Presentation)** February 2023  
University of Yazd, Iran ([Link](#))
4. **National Meeting on Gravity and Cosmology** January 2023  
Sharif University of Technology, Iran
5. **11<sup>th</sup> National Meeting on Astronomy and Astrophysics** January 2018  
Ferdowsi University of Mashhad, Iran
6. **International Astrometric Search Collaboration** July-August 2018  
Hardin-Simmon University and Pan-Starrs observatory, USA
7. **Meeting on Cosmology: from theory to observations** February 2018  
Institute for Research in Fundamental Sciences (IPM), Iran
8. **Workshop on Observational Astronomy** January 2018  
Ferdowsi University of Mashhad, Iran
9. **6<sup>th</sup> National Meeting on Astronomy and Astrophysics** January 2013  
Birjand University, Iran
10. **Observational Astronomy Summer School** August 2013  
Institute for Research in Fundamental Sciences (IPM), Iran

## Awards

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Second place in the National competition of the Transit of Mercury Iran-2016

## Invited Talks

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1. Astronomical Club of Mashhad, Iran 2011-2013
2. Khayyam University, Curiosity landing on Mars, Mashhad, Iran August 2012
3. Khayyam University, World Space Week, Mashhad, Iran 2013
4. Academy of Sciences and Astronomy of Mashhad, World Astronomy Day, World Physics Day, Mashhad, Iran 2010-2013
5. Khayyam University of Mashhad, Introduction of Astro Photometry, Mashhad, Iran 2014
6. Khayyam University of Mashhad, Astrophotography, Mashhad, Iran 2017

7. Society of Photography, Astrophotography, Chenaran, Iran 2017
8. Salman Higher Education Institute, A Day with Astronomy, Mashhad, Iran 2018
9. University of Zanjan, Astronomers meeting, Zanjan, Iran 2023

## **Languages**

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**English:** Fluent-B2 (Scheduled to take the IELTS score soon)

**Farsi:** Native

**Arabic:** Preliminary